

Percent; Ratio&Proportion

1. [March US 2023]

If $x > 0$ and $p\%$ of x is 13, which expression represents x in terms of p

- A) $13p$ C) $\frac{100p}{13}$
B) $\frac{13p}{100}$ D) $\frac{(100)(13)}{p}$

2. [March US 2023]

What is the 90% of 80?

- A) 10 C) 88
B) 72 D) 152

3. [June 2023]

A scientist analyzed a soil sample with a mass of 900 grams and determined that it contained 189 grams of water. What is the percentage of water, by mass, in this soil sample?

- A) 9% C) 18.9%
B) 9.9% D) 21%

4. [June 2023]

The ratio of green tiles to blue tile in a piece of artwork is 5 to 2. If there are 16 blue tiles in the piece of artwork, how many green tiles are there?

5. [August US 2023]

If $\frac{g}{h} = \frac{1}{100}$ and $\frac{10g}{kh} = \frac{1}{100}$,
what is the value of k ?

- A) 0.1 C) 100
B) 10 D) 1,000

6. [August US 2023]

The positive quantity r is increased by 20%, and then the resulting quantity is decreased by 25%. Which expression represents the result of these two changes?

- A) $0.90r$ C) $1.05r$
B) $0.95r$ D) $15.0r$

7. [November 2023]

Julia purchased 600 feet of fencing. She used 70% of this fencing to surround a vegetable garden. How many feet of fencing did Julia use to surround the vegetable garden?

- A) 14 C) 420
B) 35 D) 530

8. [November 2023]

The ratio 36 to m is equivalent to the ratio 2 to 12. What is the value of m ?

9. [December 2023]

A clothing store buys shirts at a wholesale price of 5.00 dollars each and resells them each at a price that is 270% of the wholesale price. At the end of the season, any remaining shirts are marked at a discounted price that is 70% of the retail price. What is the discounted price of each remaining shirt, in dollars?

10. [March 2024]

A real estate company offers a series of three webinars. 1,250 people attended the first webinar. 68% of the people who attended the first webinar attended the second webinar, and 32% of the people who attended the first and second webinars attended the third webinar. How many people attended all three webinars?

11. [March 2024]

If $\frac{2a}{b} = 5.8$ and $\frac{a}{bn} = 23.2$, what is the value of $\frac{1}{n}$?

12. [March 2024]

The speeds of particles A, B, and C are a meters per second, b meters per second, and c meters per second, respectively. If the speed of particle A is 7,900% of the speed of particle C and the speed of particle C is 0.008% of the speed of particle B, which expression represents the value of $a + b$ in terms of c ?

- A) $12.579c$ C) $7.908c$
B) $8.025c$ D) $2.040c$

13. [May 2024]

For a certain rectangular region, the ratio of its length to its width is 10 : 4. If the width of the rectangular region increases by 7 units, how must the length change to maintain this ratio?

- A) It must decrease by 7 units.
B) It must increase by 7 units.
C) It must increase by 17.5 units.
D) It must decrease by 17.5 units.

14. [May 2024]

The composition of an animal is defined as the muscles, bones, and fat of the animal. A scientist studied the composition of one young swamp buffalo and determined the buffalo had 133 kilograms of muscle, which made up approximately 65.9% of its composition. Of the remaining composition of this buffalo, approximately 46.2% was bone, and the remainder was fat. Based on these approximations, to the nearest tenth, how many kilograms of this buffalo's composition was bone?

15. [May 2024]

Raheem bought 9 shirts that were each the same price. He used a coupon for \$54 off the entire purchase. The cost for the entire purchase after using the coupon was \$108. What was the original price, in dollars, for 1 shirt?

16. [May 2024]

The combined original price for a towel and chair is \$36. After a 45% discount to the chair and a 25% discount to the towel are applied, the combined sale price for the two items is \$25. Which system of equations gives the original price t in dollars of the towel and c in dollars of the chair?

- A) $t + c = 36$
 $0.45t + 0.25c = 25$
B) $t + c = 36$
 $0.55t + 0.75c = 25$
C) $t + c = 36$
 $0.25t + 0.45c = 25$
D) $t + c = 36$
 $0.75t + 0.55c = 25$

17. [May US 2024]

After 20% of the original number of objects in a group were removed from the group, 180 objects remained in the group. How many objects were removed from the group?

- A) 36 C) 225
B) 45 D) 900

18. [May 2024]

When a collection of books is sorted, 52% are nonfiction and the remaining books are fiction. Of those that are fiction, 75% are westerns and the remaining fiction books are romance. If there are 933 romance books in this collection, what is the total number of books in the collection?

19. [May 2024]

The positive number a is 2,800% of the number c , and c is 20% of the number b . What is the value of $a - b$ in terms of c ?

- A) $23c$
- B) $140c$
- C) $560c$
- D) $2,240c$

20. [May 2024]

For each of two rectangles, the ratio of the rectangle's length to its width is 10 to 5. The width of rectangle X is 16.5 times the width of rectangle Y . How does the length of rectangle X compare to the length of rectangle Y ?

- A) The length of rectangle X is 0.5 times the length of rectangle Y .
- B) The length of rectangle X is 2 times the length of rectangle Y .
- C) The length of rectangle X is 16.5 times the length of rectangle Y .
- D) The length of rectangle X is 33 times the length of rectangle Y .

21. [June 2024]

The mass of object A is 498% of the mass of object B, and the mass of object A is 0.830% of the mass of object C. If the mass of object C is $p\%$ of the mass of object B, what is the value of p ?

22. [June 2024]

There is a proportional relationship between p and t . If $p = 2,340$ when $t = 3,120$, then what is the value of p when $t = 2,496$?

- A) 1,716
- B) 1,872
- C) 3,276
- D) 3,328

23. [August US 2024]

Which expression is equivalent to $\sqrt{x^{153}}$, where $x > 0$?

A) $x^{\frac{2}{153}}$

B) $x^{\frac{153}{2}}$

C) x^{151}

D) x^{306}

24. [August 2024]

A square map has a side length of 35 inches, and 1 inch on the map represents an actual distance of 17 miles. A smaller version of the same map is printed as a square with the side length 55% shorter than the side length of the previous map. On the smaller map, which of the following is closest to the actual distance, in miles, represented by 1 inch?

- A) 7.65
- B) 10.97
- C) 19.25
- D) 37.78

25. [August US 2024]

For the positive quantities h, j , and k . 94% of h is equivalent to 47% of j , and j is equivalent to 30% of k . What percentage of k is h ? (Disregard the % sign when entering your answer. For example, if you answer is 39%, enter 39)

26. [August Int 2024]

A company studied a random sample of 20 lawns in a small town. The company found that 4 of these lawns contain fertilizer. If there are 260 lawns in the town, which of the following is the best estimate of the number of lawns that contain fertilizer?

- A) 3
- B) 5
- C) 13
- D) 52

27. [August 2024]

Each side of square A has a length of 24 inches. Each side of square A is multiplied by a scale factor of 2 to create square B . What is the length, in inches, of each side of square B ?

- A) 22
- B) 24
- C) 26
- D) 48

28. [August 2024]

The table summarizes the number of objects in each group.

Group	Number of objects
A	325
B	68
C	572
D	102
Total	1067

The number of objects in group C is $p\%$ of the number of objects in group A . What is the value of p ?

29. [August 2024]

A clothing store buys shirts at a wholesale price of 4.00 dollars each and resells them each at a retail price that is 280% of the wholesale price. At the end of the season, any remaining shirts are marked at a discounted price that is 80% off the retail price. What is the discounted price of each remaining shirt, in dollars?

30. [August US 2024]

A clothing store buys shirts at a wholesale price of 7.00 dollars each and resells them each at a retail price that is 260% of the wholesale price. At the end of the season, any remaining shirts are marked at a discounted price that is 75% off the retail price. What is the discounted price of each remaining shirt, in dollars?

31. [June 2024]

The mass of object A is 483% of the mass of object B , and the mass of object A is 0.069% of the mass of object C . If the mass of object C is $p\%$ of the mass of object B , what is the value of $\frac{p}{1,000}$?

32. [October US 2024]

The positive number a is 3,700% of the number c , and c is 20% of the number b . If $a - b = wc$, where w is a constant, what is the value of w ?

33. [October 2024]

A clothing store buys shirts at a wholesale price of 5.00 dollars each and resells them each at a retail price that is 270% of the wholesale price. At the end of the season, any remaining shirts are marked at a discounted price that is 70% off the retail price. What is the discounted price of each remaining shirt, in dollars?

34. [October 2024]

The positive number a is 2,352% of the sum of the positive numbers b and c , and b is 84% of c . What percent of b is a ?

- A) 5,152% C) 51.52%
B) 2,436% D) 24.36%

35. [October US 2024]

The composition of an animal is defined as the muscles, bones, and fat of the animal. A scientist studied the composition of one young swamp buffalo and determined that the buffalo had 139.2 kilograms of muscle, which made up approximately 67.7% of its composition. Of the remaining composition of this buffalo, approximately 51.4% was bone, and the remainder was fat. Based on these approximations, to the nearest tenth, how many kilograms of this buffalo's composition was bone.

36. [November 2024]

A researcher surveyed undergraduate students, graduate students, and postgraduate students. The number of undergraduate students surveyed was 7450% of the number of postdoctoral students surveyed and the number of graduate students surveyed was 24% of the undergraduate students surveyed. If there were 3576 graduate students surveyed, what was the sum of the number of undergraduate students and postdoctoral students surveyed?

37. [October US 2024]

The ratio of 12 to x is equal to the ratio of 156 to y . Which expression represents y in terms of x ?

- A) $\frac{1}{13}x$ C) $\frac{1}{12}x$
B) $13x$ D) $12x$

38. [October US 2024]

For the positive quantities h , j , and k , 27% of h is the equivalent of 24% of j , and j is equivalent to 45% of k . What percentage of k is h ? (Disregard the % sign when entering your answer. For example, if your answer is 39%, enter 39)

39. [November US 2024]

A chemist mixed x liters of a 2% saline solution with y liters of a 5% saline solution to produce a 3% saline solution. Which equation best represents this situation? (Assume the volumes of the solutions are additive.)

- A) $0.02x + 0.05y = 3(x + y)$
B) $0.02x + 0.05y = 0.03(x + y)$
C) $0.2x + 0.5y = 3(x + y)$
D) $0.2x + 0.5y = 0.3(x + y)$

40. [November US 2024]

The ratio 5 to x is equivalent to the ratio y to 8. Which equation represents x in terms of y ?

- A) $x = \frac{5y}{8}$ C) $x = \frac{y}{40}$
B) $x = \frac{8y}{5}$ D) $x = \frac{40}{y}$

41. [November US 2024]

The mass of object A is 416% of the mass of object B, and the mass of object A is 0.064% of the mass of object C. If the mass of object C is $p\%$ of the mass of object B, what is the value of $\frac{p}{1,000}$?

42. [December US 2024]

Julia purchased 800 feet of fencing. She used 60% of this fencing to surround a vegetable garden. How many feet of fencing did Julia use to surround a vegetable garden?

- A) 12 C) 480
B) 30 D) 740

43. [December US 2024]

The value of an autographed baseball increased by 178% from the end of 2012 to the end of 2013 and then decreased by 21% from the end of 2013 to the end of 2014. What was the net percentage increase in the value of the autographed baseball from the end of 2012 to the end of 2014?

- A) 119.62% C) 157.00%
B) 140.62% D) 236.38%

44. [December 2024]

On a plot of land, 52.0% of the square footage is farmland and the remaining square footage is pasture. There are buildings on exactly 21.5% of the square footage of the farmland, and there are buildings on exactly 14.0% of the square footage of the pasture. If there are buildings on exactly $p\%$ of the square footage of the plot of land, what is the value of p ?

45. [December 2024]

The speeds of particles A, B, and C are a meters per second, b meters per second, and c meters per second, respectively. If the speed of particle A is 6,000% of the speed of particle C and the speed of particle C is 0.008% of the speed of particle B, which expression represents the value of $a + b$ in terms of c ?

- A) $12,560c$ C) $6,008c$
B) $6,125c$ D) $1,850c$

46. [December 2024]

If $x > 0$ and x is 33% of y , which expression represents y in terms of x ?

A) $\frac{100}{33}x$

B) $\frac{10}{33}x$

C) $\frac{1}{33}x$

D) $\frac{67}{100}x$

A) 23

B) 33

C) 41

D) 43

47. [December 2024]

Jasmin grows bean pods in her garden. This year, she harvested 330 bean pods and saved 10% of them to plant next year. How many of the harvested bean pods did Jasmin save to plant next year?

48. [December 2024]

Levi and Marissa each have stamp collections. The number of stamps in Levi's collection is 200% of the number of stamps in Marissa's collection. If there are 460 stamps in Marissa's collection, how many stamps are in Levi's collection?

A) 230

B) 460

C) 920

D) 1,380

Questions 49 and 50 refer to the following information

	Candidate 1	Candidate 2	Other candidates	Total
County A	35,954	34,777	1,640	72,371
County B	165,973	77,837	6,862	250,672
County C	46,475	54,235	2,026	102,736
Total	248,402	166,849	10,528	425,779

For three counties, the table summarizes the number of residents who voted, by county and by candidate, during a 2016 US gubernatorial election. In 2016, the total number of residents in all three of these counties was 960,000, and the total area of these three counties was 1,949 square miles.

49. [August US 2023]

To the nearest whole percent, the total number of residents who voted for Candidate 1 is what percent of the total number of residents who voted for Candidate 2?

A) 67%

B) 86%

C) 103%

D) 149%

50. [August US 2023]

For these three counties, which of the following is closest to the number of residents who voted per square mile?

A) 650

B) 493

C) 218

D) 127